Unit 1 (1 mark question) 15 question

Question	1	2	3	4	ANSWER
When to reduce the Price	Use historical data to describe for the same product	Use historical data to describe for the different product	Use historical data to describe for any product	None of above	1
How much price to be reduced?	Retailer may use Prescriptive analytics to find the best set of pricing and advertising to maximise sales revenue	Retailer may use desriptive analytics to know the pricing	Retailer may use predictive to forcast the price	None of above	1
The retailer may use predctive models for	Predcting sales based on departmental store employees	Predcting sales based on price	Predcting sales based on inventory	All of above	2
The retailer will get help from the prescptive model	by finding the best set of pricing and advertising to maximise the production price	by finding the best set of pricing and advertising to minimise sales revenue	by finding the best set of pricing and advertising to maximise sales revenue	None of the above	3
Summarising prices, units sold, advertising, covered in	Preditive Analytics	Descriptive analytics	Prescriptive analytics	None of the above	2
How to use Business Analytics?	The grocer can develop analytical model to predict sales	The grocer can develop descriptive model to understand customer	The grocer can develop decision model to summarise customer requirement	None of above	1

The variable to be used by groccer are (Marks : 1)	Sales, Price, Coupon, and advertising	Advertising	sales	None of the above	1
The dependent variable is (Marks : 1)	Sales, Coupon, and advertising	Advertising	sales	None of above	3
The relationship between sales and advertisement (Marks : 1)	Neutral	Postive	Negative	Unpredictable	2
The relationship between Price and sales (Marks : 1)	Neutral	Postive	Negative	Unpredictable	3
Classify "Order Number" based on Data types (Marks : 1)	Interval	Ratio	Nominal	Ordinal	3
Classify "Item Number" based on Data types (Marks : 1)	Nominal	Ordinal	Interval	Ratio	1
Classify "Cost" based on Data types (Marks : 1)	Nominal	Ordinal	Interval	Ratio	3
Classify "Supplier" based on Data types (Marks : 1)	Nominal	Ordinal	Interval	Ratio	1
Classify "Arrival dates" based on Data types (Marks : 1)	Nominal	Ordinal	Interval	Ratio	3

Unit 1(Passage) 15 question

Passage	Question	1	2	3	4	ANSWER
A firm wishes to determine the best pricing for one of its products in order to maximize revenue. Analysts determined the following model: Sales = -2.9485(price) + 3240.9 Total revenue = (price)(sales)	If price is Rs. 21.25 find sales (Marks: 2)	3218.98	3178.24	2178.19	4356	2
A firm wishes to determine the best pricing for one of its products in order to maximize revenue. Analysts determined the following model: Sales = -2.9485(price) + 3240.9 Total revenue = (price)(sales)	If price is Rs. 50 find sales (Marks: 2)	2083.44	5467.8	3093.475	3095	3
A firm wishes to determine the best pricing for one of its products in order to maximize revenue. Analysts determined the following model: Sales = -2.9485(price) + 3240.9 Total revenue = (price)(sales)	Find the price for which sales is maximum (Marks: 2)	30	0	19	1	2
A firm wishes to determine the best pricing for one of its products in order to maximize revenue. Analysts determined the following model: Sales = -2.9485(price) + 3240.9 Total revenue = (price)(sales)	Find total revenue for price Rs.25 (Marks: 2)	66968.25	77768.4	86968.32	76968.313	4

A firm wishes to determine the best pricing for one of its products in order to maximize revenue. Analysts determined the following model: Sales = -2.9485(price) + 3240.9 Total revenue = (price)(sales)	Find total revenue for price Rs.55 (Marks: 2)	169330.28	179330.28	149330.28	189330.28	1
The numbers of hours worked (per week) by 400 statistics students are shown below. Interval for Number of hours worked and count of students: interval 0-9 count - 20; Interval 10-19 count - 80, interval 20 - 29 count - 200, Interval 30 - 39 count - 100.	The class width of this distribution is: (Marks: 2)	is 9	is 10	is 39, which is: the largest value minus the smallest value or 39 - 0 = 39	varies from class to class	2
The numbers of hours worked (per week) by 400 statistics students are shown below. Interval for Number of hours worked and count of students: interval 0-9 count - 20; Interval 10-19 count - 80, interval 20 - 29 count - 200, Interval 30 - 39 count - 100.	The number of students working 19 hours or less (Marks: 2)	is 80	is 100	is 180	is 300	2
The numbers of hours worked (per week) by 400 statistics students are shown below. Interval for Number of hours worked and count of students: interval 0-9 count - 20; Interval 10-19 count - 80, interval 20 - 29 count - 200, Interval 30 - 39 count - 100.	The relative frequency of students working 9 hours or less (Marks: 2)	is 20	is 100	is 0.95	0.05	4
The numbers of hours worked (per week) by 400 statistics students are shown below. Interval for Number of hours worked and count of students: interval 0-9 count - 20; Interval 10-19 count - 80, interval 20 - 29 count - 200, Interval 30 - 39 count - 100.	The percentage of students working 19 hours or less is (Marks: 2)	20%	25%	75%	80%	2

The numbers of hours worked (per week) by 400 statistics students are shown below. Interval for Number of hours worked and count of students: interval 0-9 count - 20; Interval 10-19 count - 80, interval 20 - 29 count - 200, Interval 30 - 39 count - 100.	The midpoint of the last class is (Marks: 2)	50	34	35	34.5	4
A bank developed a model for predicting the average checking and savings account balance as balance = -17,732 + 367 * age + 1,300 * years education + 0.116 * household wealth.	Suppose that a customer is 32 years old, is a college graduate (so that years education = 16), and has a household wealth of $150,000. What is the predicted bank balance? (Marks: 2)	32212	52481	65325	23564	1
A bank developed a model for predicting the average checking and savings account balance as balance = -17,732 + 367 * age + 1,300 * years education + 0.116 * household wealth.	Suppose that a customer is 32 years old, is a college graduate (so that years education = 16), and has a household wealth of $100,000. What is the predicted bank balance? (Marks: 2)	26400	26412	36254	33568	2

A bank developed a model for predicting the average checking and savings account balance as balance = -17,732 + 367 * age + 1,300 * years education + 0.116 * household wealth.	Suppose that a customer is 30 years old, is a college graduate (so that years education = 16), and has a household wealth of $150,000. What is the predicted bank balance? (Marks: 2)	25468	25879	31478	45369	3
A bank developed a model for predicting the average checking and savings account balance as balance = -17,732 + 367 * age + 1,300 * years education + 0.116 * household wealth.	Suppose that a customer is 32 years old, is a college graduate (so that years education = 16), and has a household wealth of $50,000. What is the predicted bank balance? (Marks: 2)	20612	15896	18456	21456	1
A bank developed a model for predicting the average checking and savings account balance as balance = -17,732 + 367 * age + 1,300 * years education + 0.116 * household wealth.	Suppose that a customer is 52 years old, is a college graduate (so that years education = 16), and has a household wealth of $150,000.	40123	39552	41258	42015	2

	What is the predicted bank balance? (Marks: 2)					

Unit 2 (1 mark) 40 Question

Q	1	2	3	4	A
A population consists of?	Subset of items of interest	few items of interest for a particular decision or investigation	All items of interest for a particular decision or investigation	None	3
Census is defined as	Population data	Sample data	partial data	classified data	1
A sample is a called	all items of population	subset of a population	more than population	none	2
Most data with which businesses deal are	Population	samples	incomplete	none	2
The average is formally called the	arithmetic mean	geometric mean	harmonic mean	mixed mean	1
The measure of location that specifies the middle value when the data are arranged from least to greatest is the	mean	median	mode	none	2
The observation that occurs most frequently	mean	median	mode	range	3
The degree of variation in the data, that is, the numerical spread (or compactness) of the data.	despersion	central location	association	relatedness	1
Approximately 95% of the observations will fall within	one sigma limit	two sigma limit	three sigma limit	four sigma limit	2
A standardized value, commonly called a	z-score	p-value	t-value	f-value	1
A relative measure of the dispersion in data relative to the mean and is defined as	variance	coefficient of variation (CV)	standard deviation	range	2
The degree of asymmetry of observations around the mean	coefficient of variation (CV)	standard deviation	coefficient of skewness (CS)	range	3

The foundation of statistical analysis	Sampling	analysis	population	tools	1
a description of the approach that is used to obtain samples from a population prior to any data collection activity	Tools plan	sampling plan	data plan	population	2
Subjective methods include	judgment sampling	definit sampling	finite sampling	random sampling	1
he most common probabilistic sampling approach is	judgment sampling	simple random sampling	definit sampling	finite sampling	2
a sampling plan that selects every nth item from the population	systematic sampling	judgment sampling	simple random sampling	definit sampling	1
populations that are divided into natural subsets	systematic sampling	judgment sampling	stratified sampling	simple random sampling	3
sampling is based on dividing a population into sub groups	cluster sampling	systematic sampling	judgment sampling	stratified sampling	1
The measures used to estimate population parameters;	Estimators	parameters	statistics	measures	1
a single number derived from sample data that is used to estimate the value of a population parameter.	baised estimator	point estimator	unbiased estimator	none	2
If the expected value of an estimator equals the population parameter it is intended to estimate, the estimator is said to be	unbiased	parameters	statistics	biased	1
An estimate provides a range for a population characteristic based on a sample.	Interval estimate	Point estimator	unbiased estimate	none	1
A range of values between which the value of the population pa-rameter is believed to be, along with a probability that the interval correctly estimates the true (unknown) population parameter	level of significance	confidence interval	Interval estimate	Point estimator	2

The technique focuses on drawing conclusions about populations from samples	statistical inference	technical inference	samplng inference	none	1
drawing conclusions about the value of the parameters of one or more popu- lations based on sample data	sampling	unbiased estimator	hypothesis testing	point estimator	3
drawing inferences about two contrasting propositions relating to the value of one or more population parameters, such as the mean, propor\tion, standard deviation, or variance.	Hypothesis testing	sample testing	simple testing	Analytical testing	1
the existing theory or a belief that is accepted as valid unless strong statistical evidence exists to the contrary.	alternate hypothesis	null hypothesis	simple hypothesis	mixed hypothesis	2
the complement of the null hypothesis it must be true if the null hypothesis is false	alternate hypothesis	null hypothesis	simple hypothesis	mixed hypothesis	1
the risk of drawing an incorrect conclusion when the assumed hypothesis is actually true	confidence level	one level	level of significance	none	3
The null hypothesis is actually true, but the hypothesis test incorrectly rejects it	Type I error	Type II error	null hypothesis	altyernative hypothesis	1
The null hypothesis is actually false, but the hypothesis test incorrectly fails to reject it	Type I error	Type II error	null hypothesis	altyernative hypothesis	2
The probability of making a Type I error, that is, P(rejecting H0 · H0 is true), is de- noted by a and is called the	Type I error	Type II error	level of significance	comfidence level	3
The value 1 - beta is called the	power of the test	level of significance	comfidence level	none	1

Many important business measures, such as market share or the fraction of deliveries re- ceived on time, are expressed as	Value	proportions	ratio	none	2
Two-sample test for means, variance known	z-test two sample for means	t-test two sample for means	p-value	none	1
Techniques for doing this range from plotting data on charts, extracting data from databases, and manipulating and summarizing data.	Descriptive	diagnostic	Visualisation	none	3
the process of displaying data (often in large quantities) in a meaningful fashion to provide insights that will support bet- ter decisions.	Data Visualisation	Data predition	Data sampling	None	1
a visual representation of a set of key business measures	Charts	Summary	Table	Dashboard	4
The data visualisation software widly used is	Tableau	Excel	R	Python	1

Unit 2(Passage) 15 question

Passage	**Question**	**1**	**2**	**3**	**4**	**A**

A marketing study of 800 adults in the 18–34 age group reported the following information: 1. spent less than $100 but more than $0 on children's clothing per year: 50 responses 2. spent $100–$499.99 on children's clothing per year: 275 responses 3. spent $500–$999.99 on children's clothing per year: 175 responses 4. spent nothing: the remainder	Relative frequency for less than $100	0.0225	0.0525	0.0625	0.06	3
A marketing study of 800 adults in the 18–34 age group reported the following information: 1. spent less than $100 but more than $0 on children's clothing per year: 50 responses 2. spent $100–$499.99 on children's clothing per year: 275 responses 3. spent $500–$999.99 on children's clothing per year: 175 responses 4. spent nothing: the remainder	Relative frequency for less than $500	0.40625	0.20625	0.30625	0.50625	1
A marketing study of 800 adults in the 18–34 age group reported the following information: 1. spent less than $100 but more than $0 on children's clothing per year: 50 responses 2. spent $100–$499.99 on children's clothing per year: 275 responses 3. spent $500–$999.99 on children's clothing per year: 175 responses 4. spent nothing: the remainder	Relative frequency for between $100-499.99	0.54375	0.34375	0.64375	0.44375	2
A marketing study of 800 adults in the 18–34 age group reported the following information: 1. spent less than $100 but more than $0 on children's clothing per year: 50 responses 2. spent $100–$499.99 on children's clothing per year: 275 responses 3. spent $500–$999.99 on children's clothing per year: 175 responses 4. spent nothing: the remainder	Relative frequency for between $500 - $999.99	0.21875	0.11875	0.22275	0.22225	1

A marketing study of 800 adults in the 18–34 age group reported the following information: 1. spent less than $100 but more than $0 on children's clothing per year: 50 responses 2. spent $100–$499.99 on children's clothing per year: 275 responses 3. spent $500–$999.99 on children's clothing per year: 175 responses 4. spent nothing: the remainder	Relative frequency for spent nothing	0.275	0.475	0.775	0.375	4
Computing a Confidence Interval with Unknown Population Standard Deviation	When mean = 12630, standard deviation = 5393, sample size n = 25	CI UI = 14900, LI = 11404	CI UI = 15900, LI = 12404	CI UI = 14856, LI = 10404	CI UI = 15000, LI = 12004	3
Computing a Confidence Interval with Unknown Population Standard Deviation	When mean = 12630, standard deviation = 5393, sample size n = 35	CI UI= 14483 LI= 10777	CI UI= 14600 LI= 12000	CI UI= 14400 LI= 10000	CI UI= 10083 LI= 10666	1
Computing a Confidence Interval with Unknown Population Standard Deviation	When mean = 12630, standard deviation = 5393, sample size n = 30	CI UI= 15644 LI= 12616	CI UI= 14888 LI= 10111	CI UI= 15888 LI= 14111	CI UI= 14644 LI= 10616	4
Computing a Confidence Interval with Unknown Population Standard Deviation	When mean = 12630, standard deviation = 5393, sample size n = 50	CI UI= 14563 LI= 11597	CI UI= 12163 LI= 10097	CI UI= 14163 LI= 11097	CI UI= 12100 LI= 10000	3
Computing a Confidence Interval with Unknown Population Standard Deviation	When mean = 12630, standard deviation = 5393, sample size n = 20	CI UI= 15000 LI= 10000	CI UI= 15154 LI= 10106	CI UI= 15011 LI= 10011	CI UI= 15023 LI= 10032	2
Your college or university wishes to obtain reliable information about student perceptions of administra tive communication. Describe how to design a sampling plan for this situation based on your knowledge of the structure	How would you implement simple random sampling?	selecting all odd values	selectiong all even values	selecting value using random table	none	3

and organization of your college or university.						
Your college or university wishes to obtain reliable information about student perceptions of administra tive communication. Describe how to design a sampling plan for this situation based on your knowledge of the structure and organization of your college or university.	How would you implement stratified sampling?	Selecting values based on geographical locations	selecting observation based on interest	selecting all odd values	selectiong all even values	1
Your college or university wishes to obtain reliable information about student perceptions of administra tive communication. Describe how to design a sampling plan for this situation based on your knowledge of the structure and organization of your college or university.	How would you implement cluster sam pling for this study?	Selecting values based on geographical locations	selecting all odd values	selectiong all even values	selecting observation based on interest	4
Your college or university wishes to obtain reliable information about student perceptions of administra tive communication. Describe how to design a sampling plan for this situation based on your knowledge of the structure and organization of your college or university.	What would be the pros of using simple random sampling?	Easy to implement	less data requirement	not much costly	None	1
Your college or university wishes to obtain reliable information about student perceptions of administra tive communication. Describe how to design a sampling plan for this situation based on your knowledge of the structure and organization of your college or university.	What would be the pros of using startified sampling?	to represent data evenly across all stratum	Good for large data	Can be implemented easily	Simplest approach	1

Unit 2 passage 15 question

Q 1 2 3 4 A

A community health-status survey obtained the following demographic information from the respondents: Age: 18 to 29, 30 to 45, 46 to 64, and 65 + Frequency: 297, 743, 602, 369	The cumulative frequency upto age 64	1642	2011	2020	1742	1
A community health-status survey obtained the following demographic information from the respondents: Age: 18 to 29, 30 to 45, 46 to 64, and 65 + Frequency: 297, 743, 602, 369	The total count is	2020	1700	2011	1100	3
A community health-status survey obtained the following demographic information from the respondents: Age: 18 to 29, 30 to 45, 46 to 64, and 65 + Frequency: 297, 743, 602, 369	Relative frequency for the interval 65+ is	0.1134	0.1834	0.1534	0.1111	2
A community health-status survey obtained the following demographic information from the respondents: Age: 18 to 29, 30 to 45, 46 to 64, and 65 + Frequency: 297, 743, 602, 369	The Cumulative frequency upto age 29	297	743	1040	2000	1
A community health-status survey obtained the following demographic information from the respondents: Age: 18 to 29, 30 to 45, 46 to 64, and 65 + Frequency: 297, 743, 602, 369	Relative frequency for the interval 18-29 is	0.1111	0.1476	0.1564	0.01476	2
A marketing researcher surveyed 92 individuals, ask- ing them if they liked a new product concept or not. The results are shown below: Male (Yes) - 30 Male (No) - 50 Female (Yes) - 6 Female (No) - 6	Percentage of people liked the product from male	37.50%	32.60%	42.60%	39.50%	1
A marketing researcher surveyed 92 individuals, ask- ing them if they liked a new product concept or not. The results are shown below: Male (Yes) - 30 Male (No) - 50 Female (Yes) - 6 Female (No) - 6	Proportion of the male not liked the product	64%	54.34%	44.34%	None	2

A marketing researcher surveyed 92 individuals, ask- ing them if they liked a new product concept or not. The results are shown below: Male (Yes) - 30 Male (No) - 50 Female (Yes) - 6 Female (No) - 6	Proportion of the female did not liked the product	50%	12%	6.50%	20%	3
A marketing researcher surveyed 92 individuals, ask- ing them if they liked a new product concept or not. The results are shown below: Male (Yes) - 30 Male (No) - 50 Female (Yes) - 6 Female (No) - 6	Percentage of the people liked the product	39%	29%	49%	19%	1
A marketing researcher surveyed 92 individuals, ask- ing them if they liked a new product concept or not. The results are shown below: Male (Yes) - 30 Male (No) - 50 Female (Yes) - 6 Female (No) - 6	Percentage of the people not liked the product	50.87%	60.87%	40.87%	30.87%	2
For our experiment, we know that the variance of the pop- ulation is 8.33 (because the values were uniformly distrib- uted). Therefore, the standard deviation of the population is "Standard deviation of population" = 2.89. We may compute the standard error of the mean for each of the sample sizes in our experiment using formula Standard Error of the Mean = (Population var) / (sqrt n)	For n= 10 Standard error of mean is	0.914	0.816	0.999	0.789	1
For our experiment, we know that the variance of the pop- ulation is 8.33 (because the values were uniformly distrib- uted). Therefore, the standard deviation of the population is "Standard deviation of population" = 2.89. We may compute the standard error of the mean for each of the sample sizes in our experiment using formula Standard Error of the Mean = (Population var) / (sqrt n)	For n= 20 Standard error of mean is	0.56	0.65	0.86	0.35	2

For our experiment, we know that the variance of the pop- ulation is 8.33 (because the values were uniformly distrib- uted). Therefore, the standard deviation of the population is "Standard deviation of population" = 2.89. We may compute the standard error of the mean for each of the sample sizes in our experiment using formula Standard Error of the Mean = (Population var) / (sqrt n)	For n= 25 Standard error of mean is	0.67	0.47	0.77	0.577	4
For our experiment, we know that the variance of the pop- ulation is 8.33 (because the values were uniformly distrib- uted). Therefore, the standard deviation of the population is "Standard deviation of population" = 2.89. We may compute the standard error of the mean for each of the sample sizes in our experiment using formula Standard Error of the Mean = (Population var) / (sqrt n)	For n= 100 Standard error of mean is	0.289	0.49	0.39	0.678	1
For our experiment, we know that the variance of the pop- ulation is 8.33 (because the values were uniformly distrib- uted). Therefore, the standard deviation of the population is "Standard deviation of population" = 2.89. We may compute the standard error of the mean for each of the sample sizes in our experiment using formula Standard Error of the Mean = (Population var) / (sqrt n)	For n= 500 Standard error of mean is	0.229	0.129	0.339	0.555	2

Unit 3 (1 mark) 40 question

Q	1	2	3	4	A
The task of modeler is to select or build an appropriate model that best represents the behaviour of	Real Situation	Predetable Situation	Uncontrolled situation	All of these	1
Modeling demand, supply with price is defined as	Demand flexibility	supply flexibility	Price elasticity	Price flexibility	3

The model D = a - bP; where D = Demand, P is unit price, a = Constant when P is zero and b = slope of the demand function is called as	Unbiased model	biased model	Polynomial model	Linear model	4
The model D = c(P)^(-d) is called as	Polynomial model	Linear model	Nonlinear model	None	3
Linear and nonlinear model predict	simillarly	differently	jointly	None	2
Today all data nearly captured	Manually	Face to face ineraction	Digitally	over the wire	3
The volume of data today continue to be	constant	increase	decrease	None	2
Big data sets are	unstructured	messy	both	none	3
Big data refere to massive amounts of data from	Single source	wide source	Specific source	All of these	2
Big data are captured using	Sensors	Hand held device	Manually	All of these	1
Big data help organisation to understand customer behaviour and improve	Layout of the operation	Customer Services	both	none	2
Understanding Big data requires	Basic analytical tools	Basic computer system	Advance Analytics tools	Descriptive tools	3
Data mining is focussed on	Some understanding of data	better understandin g of the data	Presentation of the data	Visualisation of the data	2
In data mining statistical tools are	raely used	somewhat used	extremly used	not used	3
One of the best software for Data mining is	Excel	SAS	SPSS	Python	2
Data mining field is	rapidly growing	stagnate	growing slowly	none	1
Techniques extensively used in data mining	data visualization, data summarizatio n	PivotTables, correlation	regression analysis, and other techniques,	All of these	4
Data mining can be considered part	descriptive	prescriptive analytics	descriptive and part prescriptive analytics	None	3
Data reduction is technique of	Predticive analytics	prescriptive analytics	Data Mining	Descriptive analytics	3
Data mining is primarily	Predticive analytics	prescriptive analytics	Descriptive analytics	All of these	1
Excel provides options for including the numerical data on which charts are based within the charts is	Data labels	Field	data options	none	1

Line charts provide a useful means for displaying data over	lables	cross tab	value	time	4
For many types of data, we are interested in understanding the relative proportion of each data source to the total is	bar chart	histogram	pie chart	line chart	3
An area chart combines the features of a	pie chart with those of line chart	Line chart with those of line chart	pie chart with those of histogram chart	none	1
Scatter charts show the relationship between	three variable	two variables	single variable	more than four variables	2
Excel distinguishes between vertical and horizontal bar charts, calling	the former vertical charts and the latter horizontal charts	the former line charts and the latter bubble charts	the former column charts and the latter bar charts	none	3
Methods of describing and summarizing data using tabular, visual, and quantitative techniques.	statistical inference	opimisation	prescriptive analysis	descriptive statistics	4
a table that shows the number of observations in each of several nonoverlapping groups.	tabular data	frequency distribution	normal data	categorical data	2
Economist use data to help companies understand and predict	Populations trend	interest rate	Industry performance	All of these	4
The quantities that can change but cannot be directly controlled by the decision maker	Decsion variable	Control Variable	Uncontrolled variable	None	3
All models are based on assumptions that reflrct the modelers view of the	Expected world	Real world	Dream world	none	2
The task of modeler is to select or build an appropriate model that best represents the behaviour of	Real Situation	Predetable Situation	Uncontrolled situation	All of these	1
Modeling demand, supply with price is defined as	Demand flexibility	supply flexibility	Price elasticity	Price flexibility	3
Big data help organisation to understand customer behaviour and improve	Layout of the operation	Customer Services	both	none	2
Understanding Big data requires	Basic analytical tools	Basic computer system	Advance Analytics tools	Descriptive tools	3
Techniques extensively used in data mining	data visualization, data summarizatio n	PivotTables, correlation	regression analysis, and other techniques,	All of these	4

Data mining is primarily	Predticive analytics	prescriptive analytics	Descriptive analytics	All of these	1
The evolution of computer facilitate collection, management,, analysis and reporting of data which is often called as	Business Formulation	Business creation	Business Intelligence	Business reporting	3
Two-sample test for means, variance unknown, assumed unequal	z-test two sample for means	t-test two sample for means	p-value	none	2
Business Analytics is supported by many tools such as	Excel, SAS etc	Tables	Reporting	Data Classification	1

Unit 3(Passage) 15 question

Passage	Question	1	2	3	4	A
The fundamental purpose of analytics is to help managers solve problems and make decisions. The techniques of analytics represent only a portion of the overall problem-solving and decision-making process.	Recognising a problem consist	The statement provided for further analysis	finding a gap between what is happening and what we think should be happening	Know the problem	None	3
The fundamental purpose of analytics is to help managers solve problems and make decisions. The techniques of analytics represent only a portion of the overall problem-solving and decision-making process.	Defining the problem consist of	Finding the real problem	Finding the solution for the problem	Finding the real problem and distinguishing it from symptoms	Understanding the problem	3

The fundamental purpose of analytics is to help managers solve problems and make decisions. The techniques of analytics represent only a portion of the overall problem-solving and decision-making process.	Structuring the Problem consist	This usually involves stating goals	This usually involves stating goals and objectives, characterizing the possible decisions, and identifying any constraints or restrictions	This usually involves stating objectives,	This usually involves stating goals and objectives	2
The fundamental purpose of analytics is to help managers solve problems and make decisions. The techniques of analytics represent only a portion of the overall problem-solving and decision-making process.	Analyzing the Problem consists	Analysis involves collection of data	It invloves sampling	It involves helping managers only	Analysis involves some sort of experimentation or solution process.	4
The fundamental purpose of analytics is to help managers solve problems and make decisions. The techniques of analytics represent only a portion of the overall problem-solving and decision-making process.	Implementing the Solution consists of	Arranging the resources	translating the results of a model back to the real world.	Getting work done	none	2
A survey handed out to individuals at a major shopping mall in a small Florida city in July asked the following: • gender • age • ethnicity • length of residency • overall satisfaction with city services (using a scale of 1–5, going from poor to excellent) • quality of	Data types of "Gender"	Nominal	Ordinal	Interval	Ratio	1

schools (using a scale of 1–5, going from poor to excellent)						
A survey handed out to individuals at a major shopping mall in a small Florida city in July asked the following: • gender • age • ethnicity • length of residency • overall satisfaction with city services (using a scale of 1–5, going from poor to excellent) • quality of schools (using a scale of 1–5, going from poor to excellent)	Data types of "Age"	Nominal	Ordinal	Interval	Ratio	3
A survey handed out to individuals at a major shopping mall in a small Florida city in July asked the following: • gender • age • ethnicity • length of residency • overall satisfaction with city services (using a scale of 1–5, going from poor to excellent) • quality of schools (using a	Data types of "ethnicity"	Nominal	Ordinal	Interval	Ratio	1

scale of 1–5, going from poor to excellent)						
A survey handed out to individuals at a major shopping mall in a small Florida city in July asked the following: • gender • age • ethnicity • length of residency • overall satisfaction with city services (using a scale of 1–5, going from poor to excellent) • quality of schools (using a scale of 1–5, going from poor to excellent)	Data types of "Length of residency"	Nominal	Ordinal	Interval	Ratio	3
A survey handed out to individuals at a major shopping mall in a small Florida city in July asked the following: • gender • age • ethnicity • length of residency • overall satisfaction with city services (using a scale of 1–5, going from poor to excellent) • quality of schools (using a	Data types of "Quality of schooling"	Nominal	Ordinal	Interval	Ratio	2

scale of 1–5, going from poor to excellent)						
Four key marketing decision variables are price (P), advertising (A), transportation (T), and product quality (Q). Consumer demand (D) is influenced by these variables. The simplest model for describing demand in terms of these variables is D = k - pP + aA + tT + qQ where k, p, a, t, and q are positive constants	How does a change in Price variable affect demand?	Neutral	Postive	Negative	none	3
Four key marketing decision variables are price (P), advertising (A), transportation (T), and product quality (Q). Consumer demand (D) is influenced by these variables. The simplest model for describing demand in terms of these variables is D = k - pP + aA + tT + qQ where k, p, a, t, and q are positive constants	How does a change in Advertising variable affect demand?	Neutral	Postive	Negative	none	2

Four key marketing decision variables are price (P), advertising (A), transportation (T), and product quality (Q). Consumer demand (D) is influenced by these variables. The simplest model for describing demand in terms of these variables is D = k - pP + aA + tT + qQ where k, p, a, t, and q are positive constants	How does a change in Transporation variable affect demand?	Neutral	Postive	Negative	none	2
Four key marketing decision variables are price (P), advertising (A), transportation (T), and product quality (Q). Consumer demand (D) is influenced by these variables. The simplest model for describing demand in terms of these variables is D = k - pP + aA + tT + qQ where k, p, a, t, and q are positive constants	The equation is Linear with number of independent variable	1	2	3	4	4
Four key marketing decision variables are price (P), advertising (A), transportation (T), and product quality (Q). Consumer demand (D) is influenced by these variables. The simplest model for describing demand in terms of these variables is D = k - pP + aA + tT + qQ	The decision variable is	Price	Advertising	Demand	Transportation	3

where k, p, a, t, and q are positive constants						

Extra(Sheet 7) 15 question passage

	Q	1	2	3	4	A
Classifying Credit-Approval Decisions Intuitively: Another way of classifying the groups is to use both the credit score and years of credit history by visually drawing a straight line to separate the groups, Using a little algebra, we can calculate the equation of the line as years = -0.095 x credit score + 74.66 Therefore, we can propose a different classification rule: whenever "years + 0.095 X credit score <= 74.66"	If years = 5, Credit score = 725, The application decision will be	Application accepted	Reject application	Undecided	Insufficient data	2

reject the application						
Classifying Credit-Approval Decisions Intuitively: Another way of classifying the groups is to use both the credit score and years of credit history by visually drawing a straight line to separate the groups, Using a little algebra, we can calculate the equation of the line as years = -0.095 x credit score + 74.66 Therefore, we can propose a different classification rule: whenever "years + 0.095 X credit score <= 74.66" reject the application	If years = 15, Credit score = 825, The application decision will be	Application accepted	Reject application	Undecided	Insufficient data	1

Classifying Credit-Approval Decisions Intuitively: Another way of classifying the groups is to use both the credit score and years of credit history by visually drawing a straight line to separate the groups, Using a little algebra, we can calculate the equation of the line as years = -0.095 x credit score + 74.66 Therefore, we can propose a different classification rule: whenever "years + 0.095 X credit score <= 74.66" reject the application	If years = 0, Credit score = 825, The application decision will be	Application accepted	Reject application	Undecided	Insufficient data	1
Classifying Credit-Approval Decisions Intuitively: Another way of classifying the groups is to use both the credit score and years of credit history by visually drawing a straight line to separate the groups, Using a little algebra, we can calculate the equation of the line as years = -	If years = 10, Credit score = 625, The application decision will be	Application accepted	Reject application	Undecided	Insufficient data	2

0.095 x credit score + 74.66 Therefore, we can propose a different classification rule: whenever "years + 0.095 X credit score <= 74.66" reject the application						
Classifying Credit-Approval Decisions Intuitively: Another way of classifying the groups is to use both the credit score and years of credit history by visually drawing a straight line to separate the groups, Using a little algebra, we can calculate the equation of the line as years = -0.095 x credit score + 74.66 Therefore, we can propose a different classification rule: whenever "years + 0.095 X credit score <= 74.66" reject the application	If years = 6, Credit score = 723, The application decision will be	Application accepted	Reject application	Undecided	Insufficient data	1
Developing any Prescriptive model i.e. optimization model consists of four basic steps	The step "Identify the decision variables" define.	Decision variables are the known values that the model use	Decision variables are the values that the model compare	Decision variables are the unknown values that the model seeks to determine	None	3

Developing any Prescriptive model i.e. optimization model consists of four basic steps	The step Identify the objective function define.	The quantity we seek to minimizeonly is called the objective function	The quantity we seek to minimize or maximize is called the objective function	The quantity we seek to maximize only is called the objective function	None	2
Developing any Prescriptive model i.e. optimization model consists of four basic steps	The step " Identify all appropriate constraints." define.	Constraints are limitations, requirements, or other restrictions that are imposed on any solution, either from practical or technological considerations or by management policy.	Constraints are only management policy.	Constraints are ways to any solution, either from practical or technological considerations .	Constraints are technological considerations .	1
Developing any Prescriptive model i.e. optimization model consists of four basic steps	The step " Write the objective function and constraints as mathematical expressions" defines	The way to translate the objective function into mathematical expressions.	The way to translate the constraints into mathematical expressions.	The way to translate the objective function and constraints into mathematical expressions.	None	3
Developing any Prescriptive model i.e. optimization model consists of four basic steps	The order of all the four steps are	Decion Variable, Objective Function, Constraints, Mathematical Expression	Constraints, Decion Variable, Objective Function, Mathematical Expression	Mathematical Expression, Constraints, Decion Variable, Objective Function,	Objective Function, Mathematical Expression, Constraints, Decion Variable,	1
We apply procedure to two different types of hypothesis tests; the first involving a single population (called one-sample tests) and, later, tests involving more than one popu- lation (multiple-	The first step of the hypothesis procedure is	Identifying the population parameter of interest and formulating the hypoth- eses to test	Selecting a level of significance, which defines the risk of drawing an incor- rect conclusion when the assumed hypothesis is actually true	Determining a decision rule on which to base a conclusion	Collecting data and calculating a test statistic	1

sample tests).						
We apply procedure to two different types of hypothesis tests; the first involving a single population (called one-sample tests) and, later, tests involving more than one popu- lation (multiple-sample tests).	The Second step of the hypothesis procedure is	Determining a decision rule on which to base a conclusion	Collecting data and calculating a test statistic	Selecting a level of significance, which defines the risk of drawing an incor- rect conclusion when the assumed hypothesis is actually true	Identifying the population parameter of interest and formulating the hypoth- eses to test	3
We apply procedure to two different types of hypothesis tests; the first involving a single population (called one-sample tests) and, later, tests involving more than one popu- lation (multiple-sample tests).	The third step of the hypothesis procedure is	Collecting data and calculating a test statistic	Determining a decision rule on which to base a conclusion	Selecting a level of significance, which defines the risk of drawing an incor- rect conclusion when the assumed hypothesis is actually true	Identifying the population parameter of interest and formulating the hypoth- eses to test	2

We apply procedure to two different types of hypothesis tests; the first involving a single population (called one-sample tests) and, later, tests involving more than one popu- lation (multiple-sample tests).	The fourth step of the hypothesis procedure is	Determining a decision rule on which to base a conclusion	Selecting a level of significance, which defines the risk of drawing an incor- rect conclusion when the assumed hypothesis is actually true	Identifying the population parameter of interest and formulating the hypoth- eses to test	Collecting data and calculating a test statistic	4
We apply procedure to two different types of hypothesis tests; the first involving a single population (called one-sample tests) and, later, tests involving more than one popu- lation (multiple-sample tests).	The fifth step of the hypothesis procedure is	Applying the decision rule to the test statistic and drawing a conclusion	Selecting a level of significance, which defines the risk of drawing an incor- rect conclusion when the assumed hypothesis is actually true	Identifying the population parameter of interest and formulating the hypoth- eses to test	Collecting data and calculating a test statistic	1